

Reviewer Pack — Minimal Local Coherence Rank Correlation with Communication ...

Entry a205b849f8ac · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Local Coherence Rank Correlation with Communication Complexity

Entry ID: a205b849f8ac **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-01 02:29:42 UTC

1. Conjecture

Field A (mathematical branch): Quantum Information Theory **Field B** (complexity object): Communication Complexity

Statement:

For every n -communication protocol P , the minimal local coherence rank ($\text{mlcr}(P)$) of its associated quantum circuit is linearly correlated with its communication complexity $C(P)$, such that $\text{mlcr}(P) = \Theta(C(P))$. Equivalently, for all protocols P , if $\text{mlcr}(P) > k \cdot C(P)$, then P requires at least an exponential amount of classical communication to simulate.

Rationale (proposer's reasoning):

The concept of local coherence rank in quantum information theory measures the degree of entanglement within a sub-system, which is related to the minimum amount of classical information required to describe the state. This conjecture suggests that this quantum mechanical property can be used to predict the complexity of communication tasks, potentially providing new insights into the nature of quantum computation and communication.

Taxonomy category: quantum_information_theory (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): a3f0d8079c6b263f

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all generated protocols P with communication complexity $C(P)$, the difference $|\text{mlcr}(P) - k \cdot C(P)|$ is less than or equal to a predefined threshold for all seeds and k values tested, and falsified if any seed produces a metric greater than the threshold.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def gaussian_elimination(matrix):
    n = len(matrix)
    for i in range(n):
        max_row = i
        for j in range(i + 1, n):
            if abs(matrix[j][i]) > abs(matrix[max_row][i]):
                max_row = j
        matrix[i], matrix[max_row] = matrix[max_row], matrix[i]
        pivot = matrix[i][i]
        if pivot == 0:
            raise ValueError("Matrix is singular")
        for j in range(i, n + 1):
            matrix[i][j] /= pivot
        for j in range(n):
            if j != i:
                factor = matrix[j][i]
                for k in range(i, n + 1):
                    matrix[j][k] -= factor * matrix[i][k]
    return matrix

def local_coherence_rank(P):
    # Placeholder function to compute the local coherence rank
    # This is a dummy implementation and should be replaced with actual logic
    return random.random()

def communication_complexity(P):
    # Placeholder function to compute the communication complexity
    # This is a dummy implementation and should be replaced with actual logic
    return random.randint(1, 10)
```

```

def run_trial(seed: int) -> dict:
    random.seed(seed)

    metric_name = "mlcr_diff"
    instances_tested = 30
    n_max = 40
    conjecture_holds = True
    counterexample = ""

    for n in [5, 10, 15, 20, 30, 40]:
        if n > n_max:
            continue

        mlcr_diffs = []

        for _ in range(instances_tested):
            P = random.randint(1, 10) # Dummy protocol
            mlcr_P = local_coherence_rank(P)
            C_P = communication_complexity(P)

            if C_P == 0:
                continue

            diff = abs(mlcr_P - 2 * C_P) # Example threshold for k=2
            mlcr_diffs.append(diff)

        mean_diff = sum(mlcr_diffs) / len(mlcr_diffs)

        if any(diff > 1 for diff in mlcr_diffs): # Example threshold for k=2
            conjecture_holds = False
            counterexample = f"n={n}, max_diff={max(mlcr_diffs)}"

    return {
        "metric_name": metric_name,
        "metric_value": mean_diff,
        "instances_tested": instances_tested,
        "n_max": n_max,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys

    seeds = [int(s) for s in sys.argv[1:]] or [
        2, 3, 5, 7, 11, 13, 17, 19, 23, 29,
        31, 37, 41, 43, 47, 53, 59, 61, 67, 71,
        73, 79, 83, 89, 97, 101, 103, 107, 109, 113
    ]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

```

```

mean_value = sum(r["metric_value"] for r in results) / len(results)
std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
elif any(not r["conjecture_holds"] for r in results) and any(r["counterexample"] != "" for r in results):
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample=\"{results[0]['counterexample']}\" first_failing_seed={first_failing_seed}")
else:
    print("RESULT: INCONCLUSIVE reason=insufficient_evidence")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

{'metric_name': 'mlcr_diff', 'metric_value': 9.824386546591931, 'instances_tested': 30, 'n_max': 40,
TRIAL: {'metric_name': 'mlcr_diff', 'metric_value': 9.873772578386324, 'instances_tested': 30, 'n_max': 40,
TRIAL: {'metric_name': 'mlcr_diff', 'metric_value': 12.056910437717542, 'instances_tested': 30, 'n_max': 40,
TRIAL: {'metric_name': 'mlcr_diff', 'metric_value': 9.978423331056213, 'instances_tested': 30, 'n_max': 40,
TRIAL: {'metric_name': 'mlcr_diff', 'metric_value': 9.86036759524379, 'instances_tested': 30, 'n_max': 40,
TRIAL: {'metric_name': 'mlcr_diff', 'metric_value': 11.831256377988485, 'instances_tested': 30, 'n_max': 40,
TRIAL: {'metric_name': 'mlcr_diff', 'metric_value': 9.881771957941565, 'instances_tested': 30, 'n_max': 40,
TRIAL: {'metric_name': 'mlcr_diff', 'metric_value': 7.627697481302694, 'instances_tested': 30, 'n_max': 40,
TRIAL: {'metric_name': 'mlcr_diff', 'metric_value': 8.96274782891916, 'instances_tested': 30, 'n_max': 40,

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_413b916e.py", line 111, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^~~~~~

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The conjecture was falsified as it was shown that for at least one seed (n=40), the difference between mlcr(P) and k-C(P) exceeds the predefined thres | next: Investigate further to identify the specific protocols and conditions under which the conjecture fails, and explore potential reasons for this discrepancy.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	16809
2	preregistration	ollama_remote	glm4:latest	0	0	9230
3	novelty	ollama_remote	glm4:latest	0	0	10425
4	novelty	ollama_remote	glm4:latest	0	0	8443
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16971
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13927
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8473
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11857
9	judge	ollama_remote	glm4:latest	0	0	11942

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 108076 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/a205b849f8ac.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/a205b849f8ac.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/a205b849f8ac.tar.gz` (if generated)